

ONTAP 9.8 overview

Hardware support updates and features enhancement

Lenovo

New hardware and manageability enhancement

ONTAP 9.8 supports the DM5100F and Cisco Nexus 3232C 100 GbE switches. It also has the following enhanced manageability features:

- The ability to check and manage the network and to troubleshoot hardware problems with ThinkSystem System Manager (TSM). Click [HERE](#) to see a screenshot.

Note: Currently, these features are only provided as a preview and are only supported on the DM5000F, DM7000F, and DM7100F

- Check active NFS clients with TSM
 - Show the NFS clients that have accessed the system in the last 48 hours, including information about volumes and storage VMs.
- QoS (Quality of service) enabled by default with TSM
 - You can disable QoS or choose a custom QoS policy during the provisioning process.

Note: Storage QoS guarantees that performance of critical workloads is not degraded by competing workloads



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Lenovo Storage Manager for DM Series

Search actions, objects, and pages

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DASHBOARD

STORAGE

NETWORK

EVENTS & JOBS

PROTECTION

HOSTS

CLUSTER

Hosts

NFS Clients

0

SMB/CIFS Clients

0

SAN

0

NVMe/FC

0

Storage

Ports

Ethernet (16)

LXCA_DMS100-01

● e0H

● e0a

● e0b

● e0c

● e0d

● e1a

● e1b

● e1c

● e1d

LXCA_DMS100-02

● e0H

● e0a

● e0b

● e0c

● e0d

● e1a

● e1b

● e1c

● e1d

FC (8)

LXCA_DMS100-01

● 2a

● 2b

● 2c

● 2d

Network Interfaces

NFS (1)

SSTest

● SsUf

SMB/CIFS (1)

SSTest

● SsUf

S3 (0)

No data was found

iSCSI (0)

No data was found

FC (0)

No data was found

Storage VMs

Storage VMs (2)

● test1-nvme

● SsTest

Volumes (3)

3

LUNs (0)

0

Buckets (1)

1

Namespaces (0)

0

Data Access

Exports (2)

default (test1-nvme)

default (SsTest)

Shares (0)

No data was found

Initiator Groups (0)

No data was found

Subsystems (0)

No data was found

Data protection enhancements

- Support for root volume encryption
 - Volume Encryption can be used to protect your root volume.
- Support for moving SnapLock volumes to a destination aggregate of the same type
 - Users can use the `volume move` command to move SnapLock volumes to a destination aggregate.
- New unspecified retention period
 - Users can explicitly set the retention time for files or use the default retention period for the volume to derive the retention time. Unless you set the retention time explicitly, SnapLock will use the default retention period to calculate the retention time.
- SnapMirror Cloud
 - ONTAP now supports Snapshot copy replication to S3 object storage, powered by SnapMirror Cloud using qualified vendor applications.

File access protocol enhancements

- Encryption support between ONTAP and domain controllers
 - SMB3 connections from ONTAP to domain controllers can be encrypted.
- Improved NFSv4.1 performance with connect support
 - A connect-enabled client can have multiple TCP connections (up to 16) associated with a single NFS mount, thus obtaining higher throughput from the available network bandwidth.
- Support for NFSv4.2
 - NFSv4.2 protocol is now supported to allow access for NFSv4.2 clients.

MetroCluster enhancements

- MetroCluster IP configuration platform support
 - DM5100F, DM7100F
- A simplified interface to manage MetroCluster operations
 - Users can manage IP MetroCluster operations with TSM, including setting up IP MetroCluster sites, pairing the sites, and configuring the clusters.
- IP MetroCluster switchover and switchback with TSM
 - TSM can be used to perform every step of planned or unplanned switchover and switchback procedures for IP MetroCluster configuration.
- Support for transition from four-node MetroCluster FC to MetroCluster IP configurations

Networking and security enhancements

- Auto port placement
 - ONTAP automatically configures broadcast domains, select ports, and helps to configure logical interfaces (LIFs), VLANs, and link aggregation groups (LAGs) based on reachability and network topology detection.
- Switch CLI simplification
 - To simplify switch commands, the cluster and storage CLIs have been consolidated.
- Virtual IP (VIP) expansion
 - New fields have been added to the network, and there is now support for the border gateway protocol (BGP) `peer-group` command. This allows you to configure two additional BGP attributes for virtual IP.

Networking and security enhancements - continued

- IPv6 variable length
 - Supported IPv6 variable prefix length range has increased from 64 to 1 through 127 bits. A value of bit 128 remains reserved for virtual IP.
- Internet Protocol security (IPsec) over wire encryption
 - To ensure data is continuously secure and encrypted even while in transit, ONTAP uses the IPsec protocol. IPsec offers data encryption for all IP traffic including the NFS, iSCSI, and SMB/CIFS protocols. IPsec provides the only encryption in flight option for iSCSI traffic. When IPsec is configured, network traffic between the client and ONTAP is protected with preventive measures to combat replay and man-in-the-middle (MITM) attacks.

SAN enhancements

- Persistent ports
 - FCP persistent ports are enabled by default on All SAN Arrays (ASAs). When persistent ports are enabled, the number of active FCP LIFs serving data to the hosts remains the same, even during storage failover operations.
- Support for NVMe protocol co-existence
 - Users can configure SCSI, NAS, and NVMe protocols on the same storage virtual machine (SVM).
- Support for SnapMirror Business Continuity (SMBC)

Storage efficiency enhancements

- Temperature sensitive storage efficiency
 - Manages storage efficiently to control both hot and cold data
- Logical space reporting for SnapMirror destination volumes
 - Displays the amount of logical space used and remaining in a SnapMirror destination volume
- Fast Directory Delete
 - The speed and efficiency of directory delete has been improved, enabling directories – large directories in particular – to be deleted with lower latencies. This improved method of directory deleting can be implemented through a new REST API or by using the ONTAP CLI.

Storage resource management enhancements

- ONTAP File System Analytics for FlexGroup and FlexVol volumes
- FlexCache volumes now support the SMB protocol
- SnapMirror secondary volume as FlexCache volume
- FlexCache volume pre-population
- Asynchronous deletion of files and directories
- Volumes that are part of application provisioning are now supported
- Support for restoring pre-conversion Snapshot copies
- Tiering to ONTAP S3 object storage
- Cloud migration control
- Object tagging
- Promotion of the data to the performance tier
- Fabric Pool tiering from HDD
- QoS support for Qtrees
- Metrics for throughput floors

API enhancements

Changes to the ONTAP REST API in ONTAP 9.8

- API endpoints for new ONTAP 9.8 core features
This includes REST API support for ONTAP S3 buckets and services, SnapMirror business continuity, and file system analytics
- Security has been enhanced through the support of several services and protocols including Azure Key Vault, Google Cloud Key Management Services, IPSec, and Certificate Signing Requests.
- More efficient and modern workflows using the REST API
For example, one-click firmware updates are now available for several different types of firmware
- Performance metrics for the REST API have been expanded to include several new storage and network objects